Bayesian augmentation by chained equations for phylogenetic comparative data: the BACE R package

Daniel W. A. Noble[1], Szymon M. Drobniak[2,3], Ayumi Mizuno[4], Patrice Pottier[5], Shinichi Nakagawa[3,4]

[1] Division of Ecology and Evolution, Research School of Biology, The Australian National University, Canberra, ACT, Australia

[2] Institute of Environmental Sciences, Faculty of Biology, Jagiellonian University, Kraków, Poland

[3] School of Biological, Earth and Environmental Sciences, University of New South Wales, Sydney, NSW, Australia

[4] Department of Biological Sciences, Faculty of Science, University of Alberta, Edmonton, Alberta, Canada

[5] Department of Biological and Environmental Sciences, University of Gothenburg, Gothenburg, Sweden

Abstract

1. Missing trait data are pervasive in comparative datasets limiting the types of questions that can be tested and compromising the power of comparative analyses. Standard approaches, such as complete-case analyses and single imputation, can bias inference and underestimate uncertainty, particularly when missing data is non-random. While robust imputation methods exits, many existing software are restricted to imputing only continuous or count variables and lack the capacity to impute discrete / categorical traits (e.g., multinomial, ordered categorical), which are common trait types in comparative datasets. Such limitations have prevented the widespread use of imputation approaches in comparative biology.

2. We introduce *BACE*, an R package implementing Bayesian Augmentation by Chained Equations for phylogenetic comparative data. *BACE* enables imputation with phylogeny-informed covariance that can account for both phylogenetic and non-phylogenetic species effects, along with mixed data types, in a fully Bayesian pipeline. *BACE* provides substantial flexibility in the imputation models fit, can decouple imputation models from analysis models, and propagates both parameter and imputation uncertainty in subsequent downstream analyses.
3. We show using simulations that *BACE* recovers missing trait values with a high degree of accuracy. Across 18 different scenarios that vary in sample size (n = 100, 300, 1000 species), phylogenetic signal (Pagel's λ = 0.3, 0.7, 1) and trait correlation (ρ = 0, 0.5), we show *BACE* recovers model parameters with little bias and nominal coverage. *BACE* outperformed other comparable frequentist approaches. We illustrate *BACE* using two case studies. The first imputes amphibian heat tolerance (continuous) under standardised conditions across 5,203 species using 2,661 experimental estimates from 524 species; 89.9% of species lack experimental measurements. The second uses AVONET data for 10,563 bird species to impute migratory behaviour (ordered categorical) and hand-wing index (continuous), addressing 17 naturally missing migration records (0.16%) and testing imputation after withholding approximately 20% of observed values for each trait.
4.

Introduction

Phylogenetic comparative methods (PCMs) provide a framework for understanding how traits evolve and covary across species while accounting for their shared evolutionary history (Felsenstein, 1985; Grafen, 1989; Martins & Hansen, 1997; Pagel, 1994). Expanding trait databases, increasingly comprehensive phylogenies, and greater access to environmental data provide opportunities to investigate how evolutionary history and environmental conditions shape variation and covariation among traits allowing one to test a rich set of evolutionary hypotheses (Blomberg & Todorov, 2025; Halliwell et al., 2025). However, missing trait data limit the questions that can be addressed, and excluding species with incomplete records can reduce sample sizes, statistical power and precision (Blomberg & Todorov, 2025; Little et al., 2024; Nakagawa & Freckleton, 2008). When the availability of trait data depends on species' characteristics, this exclusion can also bias estimates of trait relationships (González-Suárez et al., 2012; Nakagawa & Freckleton, 2008). Empirical studies illustrate the consequences of missing data for answering questions in comparative biology. For example, incomplete trait coverage alters inferred relationships between mammalian traits and extinction risk (González-Suárez et al., 2012), while missing values in comparative datasets tend to be distributed nonrandomly across species — in relation to characteristics such as rarity and body size — which can bias parameter estimates and model selection when using complete-case analyses (Garamszegi & Møller, 2011).

Methods for handling missing data have a long history in statistics (Schafer & Graham, 2002).

Approaches developed for comparative datasets include phylogenetic prediction of missing trait values (Bruggeman et al., 2009; Goolsby et al., 2017), as well as multiple-imputation procedures that account for phylogenetic and species-sampling uncertainty (Blomberg & Todorov, 2025; Nakagawa & de Villemereuil, 2018). Missing values can be accommodated within model fitting, for example through Bayesian data augmentation, which samples missing values alongside model parameters (Tanner & Wong, 1987). Alternatively, missing data can be handled in a multiple-imputation workflow, where m complete datasets are generated and analysed separately. Parameter estimates and their variances are then combined, commonly using Rubin's rules, to account for both within- and between-imputation uncertainty (Noble & Nakagawa, 2021; Rubin, 1987). Data augmentation can also be used to generate the completed datasets for multiple imputation (Schafer & Graham, 2002). Multiple imputations can be generated using a joint model for all variables or separate conditional models linked through chained equations. Chained equations cycle through these models, drawing missing values for each incomplete variable using the latest values of the others. This approach, also called fully conditional specification, allows different models for continuous and categorical variables (Buuren & Groothuis-Oudshoorn, 2011). Importantly, multiple imputation — like most missing-data methods — assumes data are missing at random (MAR), and the plausibility of this assumption rests on imputation models that include the traits covarying with both the missing values and the probability of missingness (Noble & Nakagawa, 2021; Schafer & Graham, 2002). Because these relationships, and the types of the traits involved, differ from trait to trait, imputation requires substantial flexibility in the models fit to each one.

While a broad array of statistical methods and packages exist to deal with missing data, many are not suited to the datasets typical of comparative biology (Table 1). A few packages can use the phylogeny in their imputation models (e.g., *Rphylopars*, *phylomice*, *TDIP*), but most cannot (e.g., *mice*, *miceadds*, *jomo*, *pan*; Table 1). Moreover, the phylogeny-aware packages generally return a single completed dataset or single predicted values rather than multiple imputations, and most impute only continuous traits. A notable exception is *TDIP* (Gendre et al., 2024), which combines phylogenetic and machine-learning methods in an ensemble and can impute discrete traits. However, it too returns single completed datasets, and it cannot handle multi-level data structures, count data, or ordered categorical traits (Table 1). As such, there is a need for a statistically robust and flexible imputation approach that can handle multi-level data structures and a diversity of trait types.

Here, we introduce a new Bayesian imputation software package for comparative biology, named *BACE* (**B**ayesian **A**ugmentation by **C**hained **E**quations), which uses the versatile Bayesian software *MCMCglmm* (Hadfield & Nakagawa, 2010) to conduct multiple imputation through chained equations. *BACE* provides substantial flexibility in the imputation / analysis models fit, as well as a simple user interface. *BACE* also provides options to decouple imputation and analysis models. Given its Bayesian approach, imputation and analysis models can be fit jointly and a final analysis model allows for pooling of posterior distributions to account for within and across imputation uncertainty. Additionally, samples from the posterior predictive distribution can be used to generate a set of m complete datasets and these datasets can be used with any model and pooled using Rubin's rules. We describe

how *BACE* works, test its robustness using simulations, and demonstrate how to apply it in two case studies: imputing heat tolerance (continuous) under standardised conditions for 5,203 amphibian species of which only 524 (10.1%) have trait data (Pottier et al., 2025), and jointly imputing migratory behaviour (ordered categorical) and hand-wing index (continuous) for 10,563 bird species in AVONET, validating the imputations by withholding approximately 20% of the observed values of each trait.

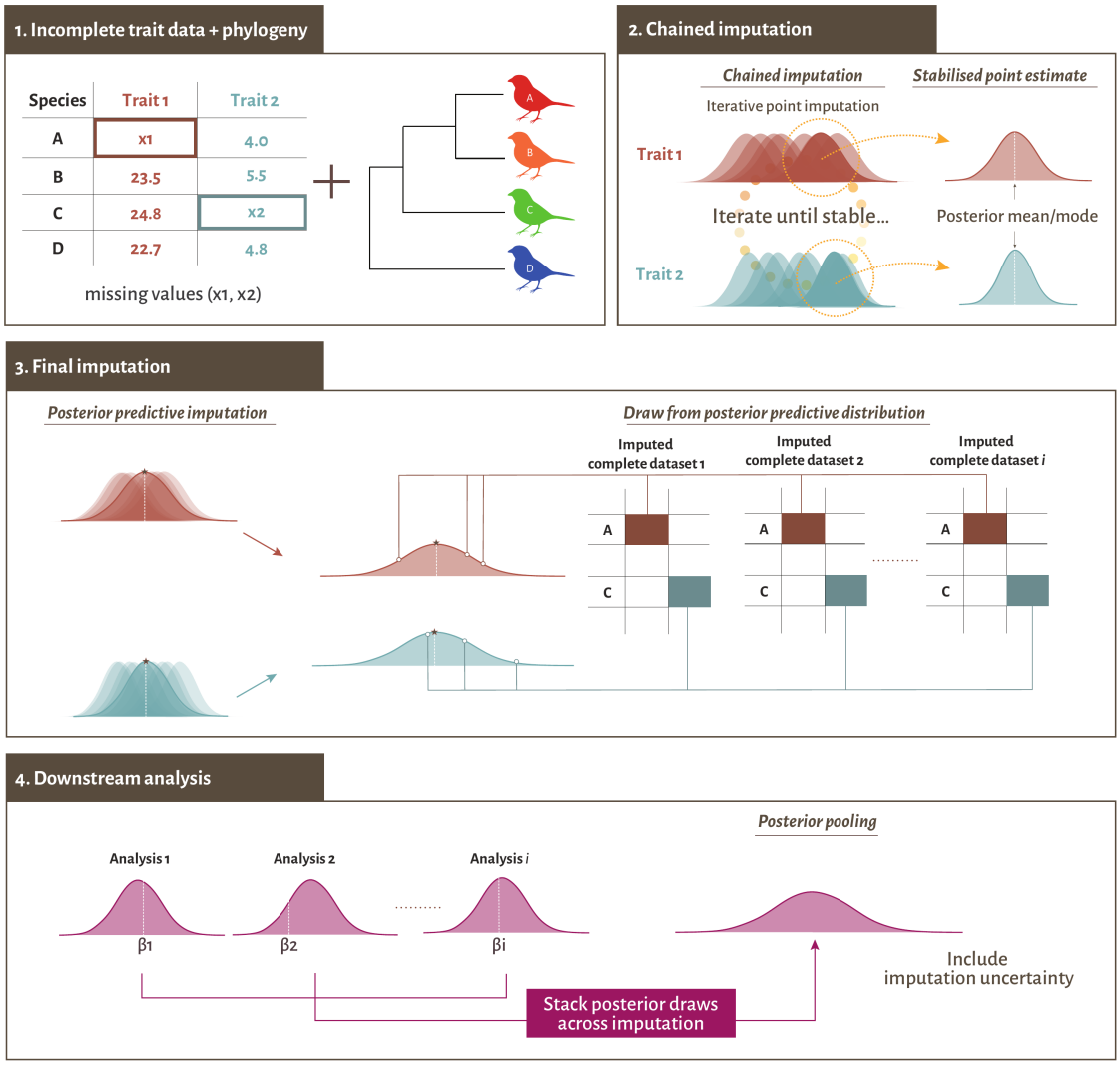


Figure 1: **Overview of the BACE imputation workflow**. (1) Inputs are an incomplete trait dataset and a phylogeny covering the species. (2) Chained imputation: a separate phylogenetic mixed model is fitted for each incomplete trait, and missing values are updated with point estimates until they stabilise. (3) Final imputation: models are refitted and missing values are drawn from their posterior predictive distributions. Repeated runs produce multiple completed datasets while retaining the observed values. (4) Downstream analysis: models are fitted to each completed dataset and posterior draws are pooled across imputations. Alternatively, parameter estimates and their variances can be combined using Rubin's rules to include imputation uncertainty.

Table 1: R tools for imputing missing data, against the needs of phylogenetic comparative analysis. *Phylogeny* (Phylo.): a tree-derived covariance informs the imputation. *Proper multiple imputation* (MI): multiple imputed datasets (or an integrated Bayesian posterior) are produced, so downstream analyses reflect imputation uncertainty. *Multilevel* (Multi.): non-independence beyond the phylogeny (random effects / clustered rows). Remaining columns: an explicit imputation model for continuous (Cont.), count, binary (Bin.), ordinal (Ord., treated as ordered) and nominal (Nom., unordered multinomial) variables. ✓ = documented in the package's reference documentation (October 2026); (✓) = partial support; ✗ = not documented.

Package	Phylo.	MI	Multi.	Cont.	Count	Bin.	Ord.	Nom.	Reference
BACE	✓	✓	(✓)[a]	✓	✓	✓	✓	✓	this paper
Rphylopars	✓	✗	(✓)	✓	✗	✗	✗	✗	Goolsby et al. (2017)
phylomice[c]	✓	✓	✗	✓	✗	✗	✗	✗	Drhlik (2016)
TDIP[c]	✓	✗	✗	✓	✗	✓	✗	✓	Gendre et al. (2024)
BHPMF[c]	(✓)[b]	✗	(✓)	✓	✗	✗	✗	✗	Schrodt et al. (2015)
mice	✗	✓	(✓)	✓	✗	✓	✓	✓	Buuren & Groothuis-Oudshoorn (2011)
miceadds	✗	✓	✓	✓	✗	✓	(✓)[d]	✗[d]	Robitzsch & Grund (2026)
micemd	✗	✓	✓	✓	✓	✓	(✓)[d]	(✓)[d]	Audigier et al. (2018)
pan	✗	✓	✓	✓	✗	✗	✗	✗	Schafer & Yucel (2002)
jomo	✗	✓	✓	✓	✗	✓	(✓)[d]	✓	Carpenter & Kenward (2013)
mitml	✗	✓	✓	✓	✗	✓	✗	✓	Grund et al. (2023)
hmi[c]	✗	✓	✓	✓	✓	✓	✓	✓	Speidel et al. (2020)
mdmb	✗	✓	(✓)	✓	✗	✓	✓	✗	Robitzsch & Lüdtke (2024)
JointAI	✗	(✓)[e]	✓	✓	✓	✓	✓	✓	Erler et al. (2021)
smcfcs	✗	✓	✗	✓	✓	✓	✓	✓	Bartlett et al. (2015)

Package	Phylo.	MI	Multi.	Cont.	Count	Bin.	Ord.	Nom.	Reference
missForest	✗	✗	✗	✓	✗	✓	✗	✓	Stekhoven & Bühlmann (2012)

[a] Phylogenetic and species (replicated taxa) random effects; other grouping factors are not supported. [b] Taxonomic hierarchy (species/genus/family), not a tree with branch lengths. [c] Not on CRAN: *phylomice* and *TDIP* (GitHub only), *BHPMF* (archived 2017), *hmi* (removed 2022). [d] Ordinal/nominal support is limited: single-level methods only (*micemd*), predictive mean matching rather than an ordinal model (*miceadds*, whose multilevel methods also cannot impute nominal variables), or substantive-model-compatible outcome models only (*jomo*). [e] Imputation is integrated with the analysis model (fully Bayesian); imputed datasets can be exported.

How *BACE* works

BACE generates multiple completed datasets from comparative data with missing trait values (Figure 1). Users provide a data frame with species identifiers, a phylogeny with branch lengths, and model formulas specifying the predictors of each incomplete trait. Continuous, count, binary, ordered categorical and unordered categorical traits can be included in the same dataset.

BACE fits a separate phylogenetic mixed model for each incomplete trait using *MCMCglmm*. Each model uses relationships among traits and shared evolutionary history to inform missing values. When species have repeated observations, a separate species effect can account for similarities not explained by the phylogeny. Imputation has two phases. First, the models are fitted repeatedly, updating missing values with point estimates. Once these estimates stabilise, *BACE* cycles through the models again, drawing plausible missing values from their posterior predictive distributions. Repeating this process produces m completed datasets, each retaining the original observed values.

The main function `bace()` runs this workflow and returns completed datasets, fitted models and diagnostics. Posterior draws from the fitted imputation models can be pooled when their parameters are of direct interest. Alternatively, users can fit a different analysis model to each completed dataset, combining posterior draws for Bayesian analyses or estimates and variances using Rubin's rules (Rubin, 1987; Zhou & Reiter, 2010). This allows imputation uncertainty to be carried into downstream analyses. *BACE* also provides convergence diagnostics to identify possible issues during model fitting. These should be checked before drawing conclusions, as some models may need more MCMC iterations or revised priors.

The *BACE* R package

Inputs, outputs, and core workflow

- Inputs: data frame, species ID, phylogeny, optional predictors, grouping factors, priors, M (number of imputations / posterior draws).
- Outputs: imputed datasets (long/wide), posterior summaries, diagnostics, helper functions for pooling and downstream analysis.

Typical user pipeline (pseudo-code)

```
# 1) Prepare data and tree (rows = taxa; factors declared; counts as integers)
# 2) Run the full workflow: chained imputation -> convergence check ->
#    n_final posterior-predictive imputations -> pooled posteriors
res <- bace(fixformula     = "y ~ x1 + x2",
            ran_phylo_form = "~ 1 | Species",
            phylo          = tree,
            data           = dat,
            runs           = 10, n_final = 50)

# 3) Diagnostics
plot(assess_convergence(res$initial_results))

# 4a) Bayesian route: pooled MCMCglmm posteriors
summary(get_pooled_model(res, "y"))

# 4b) Rubin's-rules route: any downstream model per imputed dataset
fits <- with_imputations(res, function(d) lm(y ~ x1 + x2, data = d))
pool_mi(fits)
```

Convergence diagnostics

(Structure only — content to write. See tutorial "Two kinds of convergence".) - Within-chain MCMC convergence of each per-variable model vs convergence of the chained-equations loop (the imputed-data sequence). - `assess_convergence()` criteria (%-change + trend + ACF) and the `plot_*_convergence()` family.

Pooling pathways: stacked posteriors and Rubin's rules

(Structure only. Sources: `.agents/plans/pool_mi_rubin.md`; tutorial "Interpreting the pooled output"; Zhou & Reiter 2010; BDA3 Ch. 18.) - `pool_posteriors()`: stacking per-imputation chains approximates the marginal posterior; not Rubin's rules; O(1/M) difference at small M. - `with_imputations()` + `pool_mi()`: model-agnostic Rubin's rules (Barnard–Rubin df, fmi/riv); matches `mice::pool` to machine precision. - When to use which (variance components never via Rubin).

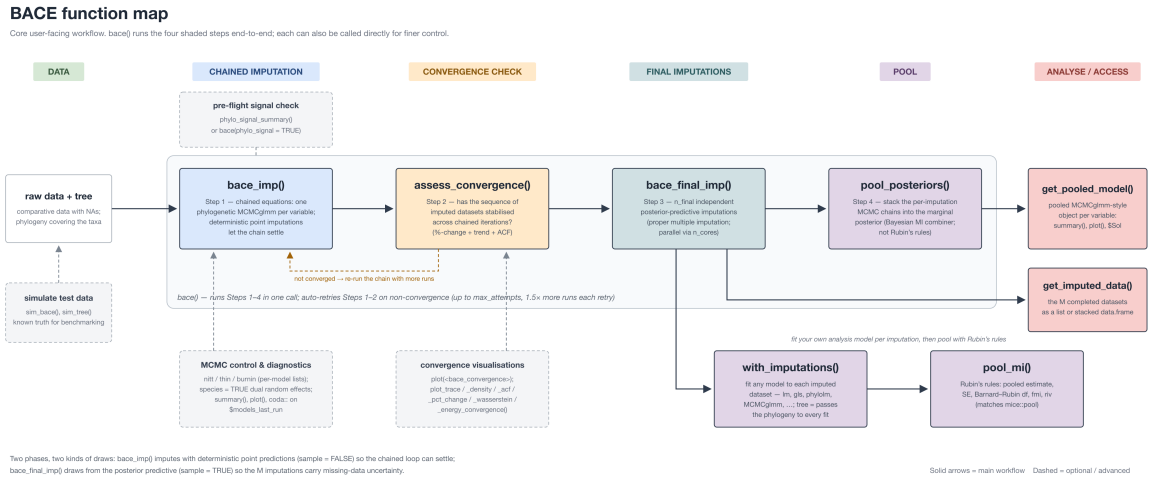


Figure 2: *(Caption to write.)* Core user-facing BACE function map.

Performance evaluation

Simulation design

(Structure only. Sources: `dev/12_recovery_simulation.R`, `dev/13_response_type_simulation.R`, `dev/simulation_results/SIMULATION_REPORT.qmd`; design details in `.agents/simulation-report.html`.) - Study A: parameter recovery (gaussian slope) under MCAR/MAR; oracle and complete-case comparators. - Study B: five response types x MCAR/MAR; per-type accuracy/coverage metrics and their finite-draw/metric ceilings (`cover95_t`, prob-based sets, Bayes ceiling for categorical).

Parameter recovery and inferential calibration (Study A)

Imputation accuracy across response types (Study B)

Benchmark against existing tools

(Pending Track B3: missForest, Rphylopars, mice as feasible. Note: pigauto is a companion package, not a competitor — see `.agents/pigauto.md` *for positioning.)*

Worked example

(Pending Track B4: AVONET at production MCMC settings; 300-species subset exists from the April benchmark.)

Discussion

(Structure only.) - What BACE adds relative to existing MI and phylogenetic-prediction tools. - Honest limitations - Practical guidance (pre-flight signal screening; when BACE is the wrong tool). - Relationship to pigauto (division of labour).

Availability and reproducibility

(Structure only.) - Package: GitHub (daniel1noble/BACE), tutorial site, CRAN (pending). - Simulation reproduction: `dev/12`, `dev/13`, `dev/14`, `SIMULATION_REPORT.qmd` (commands in `.agents/README.md`).

Acknowledgements

(To write.)

Author contributions

(To write — CRediT.)

Data and code availability

(To write.)

References

Audigier, V., White, I. R., Jolani, S., Debray, T. P. A., Quartagno, M., Carpenter, J., Buuren, S. van, & Resche-Rigon, M. (2018). Multiple imputation for multilevel data with continuous and binary variables. *Statistical Science*, *33*(2), 160–183. https://doi.org/10.1214/18-STS646

Bartlett, J. W., Seaman, S. R., White, I. R., & Carpenter, J. R. (2015). Multiple imputation of covariates by fully conditional specification: Accommodating the substantive model. *Statistical Methods in Medical Research*, *24*(4), 462–487. https://doi.org/10.1177/0962280214521348

Blomberg, S. P., & Todorov, O. S. (2025). The fallacy of single imputation for trait databases: Use multiple imputation instead. *Methods in Ecology and Evolution*, *16*(4), 658–667. https://doi.org/10.1111/2041-210X.14494

Bruggeman, J., Heringa, J., & Brandt, B. W. (2009). PhyloPars: Estimation of missing parameter values using phylogeny. *Nucleic Acids Research*, *37*(suppl_2), W179–W184. https://doi.org/10.1093/nar/gkp370

Buuren, S. van, & Groothuis-Oudshoorn, K. (2011). Mice: Multivariate imputation by chained equations in r. *Journal of Statistical Software*, *45*(3), 1–67. https://doi.org/10.18637/jss.v045.i03

Carpenter, J. R., & Kenward, M. G. (2013). *Multiple imputation and its application*. Wiley. https://doi.org/10.1002/9781119942283

Drhlik, P. (2016). *Phylomice: Phylogenetic imputation methods for mice*. https://github.com/pdrhlik/phylomice

Erler, N. S., Rizopoulos, D., & Lesaffre, E. M. E. H. (2021). JointAI: Joint analysis and imputation of incomplete data in r. *Journal of Statistical Software*, *100*(20), 1–56. https://doi.org/10.18637/jss.v100.i20

Felsenstein, J. (1985). Phylogenies and the comparative method. *The American Naturalist*, *125*(1), 1–15. https://doi.org/10.1086/284325

Garamszegi, L. Z., & Møller, A. P. (2011). Nonrandom variation in within-species sample size and missing data in phylogenetic comparative studies. *Systematic Biology*, *60*(6), 876–880. https://doi.org/10.1093/sysbio/syr060

Gendre, M., Hauffe, T., Pimiento, C., & Silvestro, D. (2024). Benchmarking imputation methods for categorical biological data. *Methods in Ecology and Evolution*, *15*(9), 1624–1638. https://doi.org/10.1111/2041-210X.14339

González-Suárez, M., Lucas, P. M., & Revilla, E. (2012). Biases in comparative analyses of extinction risk: Mind the gap. *Journal of Animal Ecology*, *81*(6), 1211–1222. https://doi.org/10.1111/j.1365-2656.2012.01999.x

Goolsby, E. W., Bruggeman, J., & Ané, C. (2017). Rphylopars: Fast multivariate phylogenetic comparative methods for missing data and within-species variation. *Methods in Ecology and Evolution*, *8*(1), 22–27. https://doi.org/10.1111/2041-210X.12612

Grafen, A. (1989). The phylogenetic regression. *Philosophical Transactions of the Royal Society of London. Series B, Biological Sciences*, *326*(1233), 119–157. https://doi.org/10.1098/rstb.1989.0106

Grund, S., Robitzsch, A., & Lüdtke, O. (2023). *Mitml: Tools for multiple imputation in multilevel modeling*. https://CRAN.R-project.org/package=mitml

Hadfield, J. D., & Nakagawa, S. (2010). General quantitative genetic methods for comparative biology: Phylogenies, taxonomies and multi-trait models for continuous and categorical characters. *Journal of Evolutionary Biology*, *23*(3), 494–508. https://doi.org/10.1111/j.1420-9101.2009.01915.x

Halliwell, B., Holland, B. R., & Yates, L. A. (2025). Multi-response phylogenetic mixed models: Concepts and application. *Biological Reviews of the Cambridge Philosophical Society*, *100*(3), 1294–1316. https://doi.org/10.1111/brv.70001

Little, R. J., Carpenter, J. R., & Lee, K. J. (2024). A comparison of three popular methods for handling missing data: Complete-case analysis, inverse probability weighting, and multiple imputation. *Sociological Methods & Research*, *53*(3), 1105–1135. https://doi.org/10.1177/00491241221113873

Martins, E. P., & Hansen, T. F. (1997). Phylogenies and the comparative method: A general approach to incorporating phylogenetic information into the analysis of interspecific data. *The American Naturalist*, *149*(4), 646–667. https://doi.org/10.1086/286013

Nakagawa, S., & de Villemereuil, P. (2018). A general method for simultaneously accounting for phylogenetic and species sampling uncertainty via rubin's rules in comparative analysis. *Systematic Biology*, *68*(4), 632–641. https://doi.org/10.1093/sysbio/syy089

Nakagawa, S., & Freckleton, R. P. (2008). Missing inaction: The dangers of ignoring missing data. *Trends in Ecology & Evolution*, *23*(11), 592–596. https://doi.org/10.1016/j.tree.2008.06.014

Noble, D. W. A., & Nakagawa, S. (2021). Planned missing data designs and methods: Options for strengthening inference, increasing research efficiency and improving animal welfare in ecological and evolutionary research. *Evolutionary Applications*, *14*(8), 1958–1968. https://doi.org/10.1111/eva.13273

Pagel, M. (1994). Detecting correlated evolution on phylogenies: A general method for the comparative analysis of discrete characters. *Proceedings of the Royal Society of London. Series B: Biological Sciences*, *255*(1342), 37–45. https://doi.org/10.1098/rspb.1994.0006

Pottier, P., Kearney, M. R., Wu, N. C., Gunderson, A. R., Rej, J. E., Rivera-Villanueva, A. N., Pollo, P., Burke, S., Drobniak, S. M., & Nakagawa, S. (2025). Vulnerability of amphibians to global warming. *Nature*, *639*(8056), 954–961. https://doi.org/10.1038/s41586-025-08665-0

Robitzsch, A., & Grund, S. (2026). *Miceadds: Some additional multiple imputation functions, especially for mice*. https://CRAN.R-project.org/package=miceadds

Robitzsch, A., & Lüdtke, O. (2024). *Mdmb: Model based treatment of missing data*. https://CRAN.R-project.org/package=mdmb

Rubin, D. B. (1987). *Multiple imputation for nonresponse in surveys*. John Wiley & Sons. https://doi.org/10.1002/9780470316696

Schafer, J. L., & Graham, J. W. (2002). Missing data: Our view of the state of the art. *Psychological Methods*, *7*(2), 147–177. https://doi.org/10.1037/1082-989X.7.2.147

Schafer, J. L., & Yucel, R. M. (2002). Computational strategies for multivariate linear mixed-effects models with missing values. *Journal of Computational and Graphical Statistics*, *11*(2), 437–457. https://doi.org/10.1198/106186002760180608

Schrodt, F., Kattge, J., Shan, H., Fazayeli, F., Joswig, J., Banerjee, A., Reichstein, M., Bönisch, G., Díaz, S., Dickie, J., Gillison, A., Karpatne, A., Lavorel, S., Leadley, P., Wirth, C. B., Wright, I. J., Wright, S. J., & Reich, P. B. (2015). BHPMF – a hierarchical bayesian approach to gap-filling and trait prediction for macroecology and functional biogeography. *Global Ecology and Biogeography*, *24*(12), 1510–1521. https://doi.org/10.1111/geb.12335

Speidel, M., Drechsler, J., & Jolani, S. (2020). The r package hmi: A convenient tool for hierarchical multiple imputation and beyond. *Journal of Statistical Software*, *95*(9), 1–48. https://doi.org/10.18637/jss.v095.i09

Stekhoven, D. J., & Bühlmann, P. (2012). MissForest—non-parametric missing value imputation for mixed-type data. *Bioinformatics*, *28*(1), 112–118. https://doi.org/10.1093/bioinformatics/btr597

Tanner, M. A., & Wong, W. H. (1987). The calculation of posterior distributions by data augmentation. *Journal of the American Statistical Association*, *82*(398), 528–540. https://doi.org/10.1080/01621459.1987.10478458

Zhou, X., & Reiter, J. P. (2010). A note on Bayesian inference after multiple imputation. *The American Statistician*, *64*(2), 159–163. https://doi.org/10.1198/tast.2010.09109